

SAM, Kazakhstan

WMO: 359250

Lat: 45.024N

Lon: 55.748E

Elev: 88

StdP: 100.27

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-21.1	-27.2	0.3	-24.1	-24.0	0.4	-20.6	11.5	-9.1	10.0	-6.1	2.3	90	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	39.4	19.3	37.7	18.8	36.1	18.5	21.5	33.3	20.5	32.3	19.6	31.8	3.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.9	25.0	16.5	11.9	25.0	15.2	10.9	24.7	62.8	33.3	59.2	31.9	56.4	31.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.3	8.5	-26.6	41.9	5.3	2.0	-30.4	43.4	-33.5	44.6	-36.4	45.7	-40.3	47.2		
(5)			-26.6	23.5	5.2	1.9	-30.3	24.9	-33.3	26.0	-36.2	27.1	-40.0	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-7.8	-6.2	2.6	12.4	19.9	25.5	27.9	26.4	18.6	10.2	1.4	-4.8		
	DBStd	13.68	6.53	6.96	5.86	5.20	4.34	3.84	3.43	3.93	4.40	4.89	5.49	5.74		
	HDD10.0	2051	551	452	234	33	1	0	0	0	1	59	261	459		
	HDD18.3	3736	809	686	486	188	34	2	0	1	48	255	509	717		
	CDD10.0	2274	0	0	6	105	307	466	555	509	261	64	2	0		
	CDD18.3	917	0	0	0	10	82	218	297	252	57	2	0	0		
	CDH23.3	12309	0	0	2	139	994	2908	4140	3375	711	41	0	0		
	CDH26.7	6684	0	0	0	39	407	1604	2418	1936	274	5	0	0		
	WSAvg	3.5	3.7	3.8	4.3	4.1	3.6	3.3	3.1	3.0	3.0	3.2	3.3	3.6		
(15) Precipitation	PrecAvg	142	10	10	12	13	20	18	12	3	5	11	15	11		
	PrecMax	193	22	31	38	38	41	54	36	13	30	31	33	23		
	PrecMin	82	0	1	0	2	5	2	1	0	0	0	2	1		
	PrecStd	30	6	7	9	9	11	15	10	4	7	10	9	7		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	8.8	20.3	30.9	35.7	40.5	41.7	41.3	36.5	27.1	16.4	8.1		
		MCWB	2.8	5.3	11.4	15.9	17.0	18.9	19.9	20.0	18.2	13.8	10.2	4.4		
	2%	DB	2.6	5.9	16.3	27.3	33.2	37.9	39.5	39.1	32.1	24.1	13.5	5.1		
		MCWB	1.6	3.8	9.7	14.9	16.4	18.4	19.5	19.4	16.0	13.1	9.0	3.3		
	5%	DB	1.3	3.5	13.7	24.3	30.9	36.1	37.9	37.1	29.6	21.6	11.2	3.1		
		MCWB	0.4	1.7	8.1	13.8	15.8	18.1	19.2	18.7	15.3	11.7	7.7	1.7		
	10%	DB	0.3	1.7	10.9	21.6	28.7	34.4	36.1	35.3	27.4	18.9	8.7	1.6		
		MCWB	-0.4	0.3	6.5	12.4	15.4	17.6	18.7	18.4	14.5	10.7	6.0	0.6		
	0.4%	WB	3.3	6.2	12.2	18.8	19.9	21.8	23.0	23.0	19.6	15.7	12.5	5.4		
		MCDB	4.2	8.0	19.1	26.0	29.2	33.7	34.3	36.1	31.1	22.2	14.7	7.3		
	2%	WB	1.7	3.8	10.1	16.3	18.2	20.3	21.6	21.2	17.4	14.1	9.8	3.6		
		MCDB	2.4	5.8	15.3	24.2	27.6	31.7	33.2	33.7	29.0	21.4	12.4	4.8		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	0.6	1.9	8.5	14.5	17.2	19.4	20.7	20.0	16.2	12.8	7.9	1.9		
		MCDB	1.2	3.2	13.2	22.2	27.4	31.3	32.4	32.4	27.1	20.0	10.6	2.8		
	10%	WB	-0.4	0.5	6.8	13.0	16.3	18.6	19.9	19.1	15.3	11.4	6.2	0.6		
		MCDB	0.3	1.5	10.6	20.6	26.3	30.7	31.9	32.5	25.5	18.0	8.8	1.5		
		MDBR	7.1	8.4	9.5	11.9	12.7	13.2	12.8	13.6	13.7	12.5	8.6	6.8		
		MCDBR	5.5	8.8	13.8	15.5	15.4	15.9	15.0	16.0	16.7	16.2	12.3	8.3		
(36) Mean Daily Temperature Range	5% DB	MCWBR	4.8	6.7	8.1	7.1	5.6	5.1	5.0	5.7	6.6	7.8	8.0	6.6		
		MCDBR	5.2	7.7	13.1	13.9	13.2	13.3	12.5	13.7	14.3	13.8	10.6	7.0		
	5% WB	MCWBR	4.7	6.1	8.1	7.3	5.8	5.1	5.3	5.8	6.5	7.4	7.6	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.384	0.457	0.509	0.502	0.484	0.469	0.453	0.419	0.416	0.382	0.371		
	taud		2.044	2.014	1.914	1.823	1.891	1.981	2.042	2.064	2.104	2.050	2.099	2.046		
	Ebn at Noon		697	763	756	754	777	794	800	797	789	717	669	634		
	Edh at Noon		105	131	166	198	191	176	163	154	136	125	99	96		
(44) All-Sky Solar Radiation	RadAvg		1.20	2.17	3.55	5.09	6.37	7.17	6.87	6.23	4.82	3.10	1.73	1.05		
	RadStd		0.23	0.38	0.35	0.46	0.44	0.43	0.28	0.34	0.28	0.20	0.22	0.19		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.20	-0.85	-1.14	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	+1.20	-0.85	-1.14	N/A	N/A	N/A	N/A

Nomenclature: See separate page